

1. Model Formulations. We investigate melt migration beneath a mid-ocean ridge using a viscous Darcy–Stokes coupled flow model [12, 6, 2]. The governing equations are derived from mixture theory and formulated in terms of the porosity, ϕ , which characterizes the volume fraction of melt. This framework captures the essential physics of a slowly deforming solid matrix, melt percolation through a porous medium, and compaction processes.

The mechanical model is coupled with transport equations for energy [5] and chemical species [10], resulting in a fully time-dependent system. At each time step, we assume local thermodynamic equilibrium, allowing the thermodynamic state to be determined from an appropriate phase diagram.

1.1. Static Flow Model. We begin by formulating the conservation laws for mass and momentum of the two-phase mixture under the Boussinesq approximation, neglecting density differences between the solid and liquid phases except the gravitational force term. The continuity equation for the mixture is given by

$$(1.1) \quad \nabla \cdot \bar{\mathbf{v}} = 0,$$

where the mixture velocity $\bar{\mathbf{v}} = \phi \mathbf{v}_l + (1 - \phi) \mathbf{v}_s$. Here, $\mathbf{v}_l$ and $\mathbf{v}_s$ denote the velocities of the liquid and solid phases, respectively, and ϕ represents the porosity. Throughout this work, subscripts l and s refer to quantities associated with the liquid (magma) and the solid matrix, respectively.

Conservation of momentum for this slowly creeping solid matrix under the assumption of low Reynolds number can be written in terms of the deviatoric stress of the mixture

$$(1.2) \quad \nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0,$$

where the deviatoric stress of the solid is

$$(1.3) \quad \boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s \left(D\mathbf{v}_s - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I} \right),$$

and $D\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient.

The mixture conservation equations (1.1) and (1.2) are insufficient to simultaneously determine the individual velocities and pressures. To close the system, additional constitutive relationships are required to describe the relative behavior of the liquid phase within the porous medium.

The relative motion between the molten magma and the deformable solid framework is modeled using a Darcy-type law, which characterizes melt segregation driven by pressure gradients and buoyancy forces within the porous medium:

$$(1.4) \quad \phi(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_\phi}{\mu_l} (\nabla p_l - \rho_f \mathbf{g}) = 0.$$

We define the relative permeability $k_\phi = k_0 \phi^{2+2\Theta}$ with a constant parameter $\Theta \in [0, 1/2]$. This relation indicates that melt segregation is regulated by the relative permeability of the porous medium. Variations in intrinsic permeability k_0 can significantly modify melt migration structures beneath mid-ocean ridges as well [9].

Another fundamental constitutive relation in mantle dynamics links the pressure difference between the liquid and solid phases to the volumetric deformation of the solid matrix in a partially molten system [6]

$$(1.5) \quad \xi_\phi \nabla \cdot \mathbf{v}_s = p_l - p_s,$$

where ξ_ϕ is often referred to as bulk viscosity or compaction viscosity. When the liquid pressure exceeds the solid pressure, the solid matrix undergoes expansion. Conversely, compaction occurs when the liquid pressure is lower than the solid pressure.

We approximate the compaction viscosity ξ_{ϕ_l} using a simplified constitutive relationship [4, 5],

$$(1.6) \quad \xi_{\phi_l} = \frac{\mu_s}{\phi_l^m}.$$

In our model, we set $m = 1$. This equation emphasize the fact that increasing melting production facilitates the compaction of the solid phase, while in the limit of no melting ($\phi_l = 0$), compaction is entirely eliminated.

When there is no melt presents $\phi = 0$, the Darcy equation (1.4) mathematically degenerates. We use the scaled formulation to handle the singularity [1]. We first define pressure potentials as

$$(1.7) \quad q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z,$$

where $g = |\mathbf{g}|$ is the magnitude of gravity acceleration and z is the depth, such that $\mathbf{g} = g \nabla z$. Scaled variables are also defined,

$$(1.8) \quad \tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r = \phi^{-\Theta} (\mathbf{v}_l - \mathbf{v}_s) \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l.$$

The problem is scaled and reformulated as

$$(1.9) \quad \tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_l} \nabla (\phi^{-1/2} \tilde{q}_l) = 0,$$

$$(1.10) \quad \mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0,$$

$$(1.11) \quad \nabla q - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) = -(1-\phi)(\rho_s - \rho_l) \mathbf{g},$$

$$(1.12) \quad \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_l - \phi^{1/2} q) = 0,$$

where the new scaled variables maintain numerical stability in the no melting ($\phi = 0$) regions [1].

1.2. Transport Model. The chemical compositions of mantle rocks and magma are complicated and challenging to track in a fully resolved multicomponent framework. Therefore, we adopt a simple two-species, two-phase model. In this framework, we neglect detailed description of the chemical reaction process and do not account for the variations in Gibbs free energy [11, 4, 5, 3].

Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state, while remaining fully miscible in the liquid phase. We extend the Boussinesq approximation, we consider a homogenous density ρ for both the solid phase and the liquid phase. Figure 1.1 illustrates the coexistence of species and phases in representative volume element. The advection-diffusion equation governing transport of the mixture chemical species are solved at each time step. We track only one species and omit the subscript i .

$$(1.13) \quad \frac{\partial \bar{c}}{\partial t} + \nabla \cdot (\bar{c} \mathbf{v} - \phi \mathcal{D}_l \nabla c_l) = 0,$$

where $\bar{c} = \sum_{j \in \{s,l\}} \phi_j c_j$ and $\bar{c} \mathbf{v} = \sum_{j \in \{s,l\}} \phi_j c_j \mathbf{v}_j$ are the mixture quantities. The diffusion term $\frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l$ is omitted in this work due to the fact that the porosity ϕ remains small ($\phi \ll 1$)

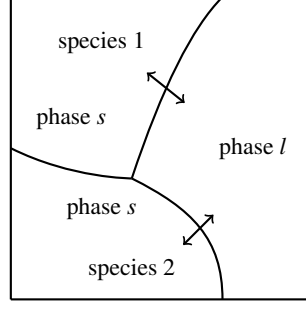


FIG. 1.1. Phases and species in an representative volume element (RVE).

in the partially molten mantle and the associated diffusivity coefficient $\mathcal{D}_l$ is comparatively negligible [10] (on the order of 10^{-10} to 10^{-9} m^2/s). Effective velocity for the mixture velocity is applied for the equilibrium transport here:

$$(1.14) \quad \mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i},$$

where the partition coefficient $D_i = c_{i,s}/c_{i,l}$. In the no melting regions, where $\phi_l = 0$, we retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. Conversely, in the case of melting completely, where $\phi_l = 1$, we obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Therefore, the definition of effective velocity is well-defined in these two limiting scenarios. We rewrite the transport equation (1.13) with effective velocity (1.14) as

$$(1.15) \quad \frac{\partial C}{\partial t} + \nabla \cdot \mathbf{v}_{\text{eff}} C = 0,$$

where we rewrite $C = \bar{c}$ for simplicity and drop the diffusion term.

We write the temperature formulation of the enthalpy transport equation under the extended Boussinesq approximation in terms of the temperature T and the latent heat L as

$$(1.16) \quad \frac{\partial \bar{H}}{\partial t} + \nabla \cdot (\bar{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot ((1 - \phi) \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_p}{c_p} T \bar{\mathbf{v}} \cdot \mathbf{g},$$

where c_p is the specific heat capacity, α_p is thermo expansion coefficient. $\bar{H} = \bar{\rho} \bar{h} / \rho c_p = \bar{h} / c_p$ is the bulk specific enthalpy and $\kappa = \bar{K} / \rho c_p$ is the bulk heat diffusivity. The last term involving gravitational acceleration is referred to as adiabatic cooling or decompression cooling term [7].

1.3. Phase Model. With the transport equations specified, we are now able to update the distribution of chemical species and energy within the system. The thermodynamic state is then determined under the assumption of local thermodynamic equilibrium at each time step.

Given that mid-ocean ridges develop in relatively low-temperature mantle environments, eutectic melting provides an appropriate conceptual framework for modeling phase transitions. In particular, the presence of a secondary component depresses the melting temperature of the mixture in the binary eutectic model.

The binary eutectic system has been widely employed in studies of chemical equilibrium across a range of geophysical and materials science applications. In mantle dynamics, it provides a tractable framework for modeling multicomponent melting processes [8]. Similar formulations have been used to describe alloy solidification in the alloy systems [13], as well as phase behavior in water–salt–brine systems in glaciers [3].

We consider a partially molten system composed of two primary components: Olivine and Orthopyroxene (opx). These components are assumed to be highly miscible in the liquid phase but immiscible in the solid phase. Olivine constitutes the dominant component of the system, whereas orthopyroxene represents the minor component. In this work, we track the concentration of the minor component orthopyroxene. A typical eutectic phase diagram is

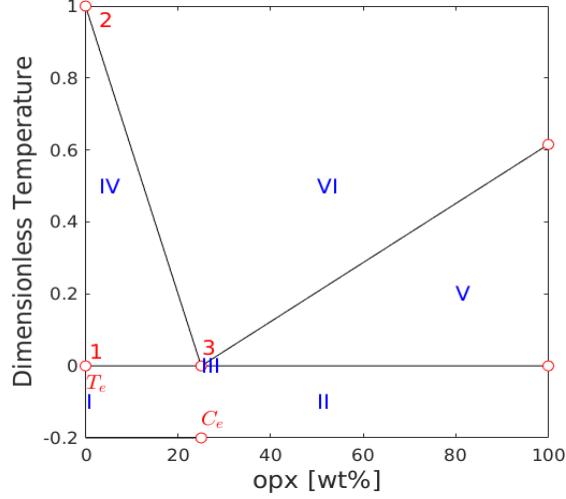


FIG. 1.2. Eutectic phase diagram of a Olivine-Orthopyroxene system.

shown in Figure 1.2. There are two key concepts about the diagram:

- The eutectic melting point or eutectic temperature T_e , represents the minimum temperature at which melting may occur in this binary system. Melting that takes place at the eutectic temperature is referred to as the eutectic melting. The temperature of the system is fixed at the eutectic temperature when eutectic melting happens.
- The eutectic bulk composition, C_e corresponds to the fixed orthopyroxene–olivine concentration ratio of the liquid generated during the eutectic melting. This composition is uniquely determined by the mixture system and remains constant as long as melting occurs at the eutectic temperature.

Pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate the solidus melting temperature T_m and the eutectic melting temperature T_e at a given pressure using the approximated Clausius–Clapeyron relationship.

$$(1.17) \quad \begin{aligned} T_m &= T_{m0} + v^{-1}(P - P_0), \\ T_e &= T_{e0} + v^{-1}(P - P_0), \end{aligned}$$

where v is the Clausius–Clapeyron constant, and P_0 is the reference pressure usually set to be 0. The melting point T_{m0} and eutectic point T_{e0} refer to their respective values at standard atmospheric pressure.

Binary eutectic melting is usually highly-nonlinear due to the strong coupling between temperature, composition, and phase proportions. To simplify the analysis, we adopt a geometric parameterization of the binary eutectic phase diagram that permits analytical evaluation. In particular, we approximate the eutectic phase boundaries using linear relations,

thereby enabling a clear distinction among the different phase regions and easy phase split calculation.

The eutectic composition of opx is set as $C_e \approx 0.25$. For illustration only, we plot bulk composition against a renormalized dimensionless temperature $\tilde{T} - T_e/\Delta T$, where $\tilde{T}_m - T_e/\Delta T = 1$ and $\tilde{T}_e - T_e/\Delta T = 0$.

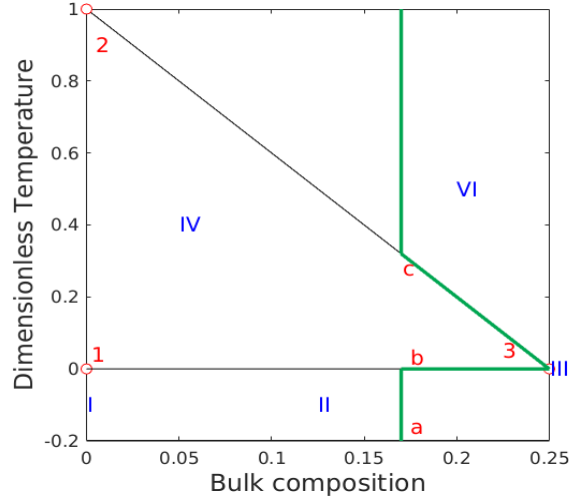


FIG. 1.3. Phase diagram when the composition of opx is smaller than the eutectic composition. Region V does not appear.

We identify the following 6 regions with different phase assemblages:

1. Single-phase sub-solidus : Olivine;
2. Two-phase sub-solidus : Olivine + Orthopyroxene;
3. Three-phase eutectic : Olivine + Orthopyroxene + melt;
4. Two-phase super-eutectic : Olivine + melt;
5. Two-phase super-eutectic : Orthopyroxene + melt;
6. Single-phase super-liquidus : melt.

In the partially molten system considered here, the bulk composition of orthopyroxene (opx) is assumed to remain below the eutectic composition. Consequently, our analysis is restricted to the region of the phase diagram Figure 1.2, where the bulk composition is less than the eutectic composition, which is highlighted in the Figure 1.3.

Melting in the eutectic system is illustrated by the green trajectory in the Figure 1.3. Consider a representative system with a bulk composition of 17% orthopyroxene (opx) that undergoes progressive heating. Initially, the system is entirely solid (Point a), and its temperature increases linearly with increasing enthalpy.

When the temperature reaches the eutectic temperature (Point b), eutectic melting begins. At this stage, both components melt simultaneously in a fixed proportion corresponding to the eutectic composition (25% opx and 75% olivine). The temperature remains constant at the eutectic value during this process.

Once all solid opx has been consumed, the system departs from the eutectic point. Further enthalpy input leads to melting of the remaining solid olivine, and the state evolves along the liquidus line (from Point b toward Point c). After complete melting of the residual olivine, the system becomes fully liquid, and the temperature again increases approximately linearly with additional enthalpy.

The melting path described above is reversible under equilibrium conditions, yielding the corresponding trajectory for equilibrium crystallization.

Quantity	Value	Units
ρ	3000	$\text{kg} \cdot \text{m}^{-3}$
ρ_r	600	$\text{kg} \cdot \text{m}^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$
L	5×10^5	$\text{J} \cdot \text{kg}^{-1}$
μ_l	1	Pascal $\cdot$ s
μ_s	10^{19}	Pascal $\cdot$ s
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	Pascal $\cdot$ K $^{-1}$

TABLE 1.1
Flow Mechanics and Thermodynamics Parameters

1.4. Parameters.

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